Moving Average Analysis

```python
df['MA_7'] = df['Close'].rolling(window=7).mean()
df['MA_30'] = df['Close'].rolling(window=30).mean()

plt.figure(figsize=(12,5))
plt.plot(df['Close'], label='Close Price')
plt.plot(df['MA_7'], label='7-Day MA')
plt.plot(df['MA_30'], label='30-Day MA')

plt.legend()
plt.title("Moving Averages")
plt.show()
```

Stationarity Check (ADF Test)

```python
result = adfuller(df['Close'].dropna())

print("ADF Statistic:", result[0])
print("p-value:", result[1])

if result[1] < 0.05:
    print("Data is Stationary")
else:
    print("Data is Non-Stationary")
```

Make Data Stationary (if required)

```python
df['Close_diff'] = df['Close'].diff()

result = adfuller(df['Close_diff'].dropna())

print("New p-value:", result[1])
```

Seasonal Decomposition

```
decomposition = seasonal_decompose(
    df['Close'],
    model='additive',
    period=30
)
decomposition.plot()
plt.show()
```

This identifies:

Trend

Seasonality

Residuals

Build Forecasting Model (ARIMA)

```python
model = ARIMA(df['Close'], order=(5,1,0))

model_fit = model.fit()

print(model_fit.summary())
```

Forecast Next 30 Days

```python
forecast = model_fit.forecast(steps=30)

print(forecast)
```

Forecast Plot

```python
plt.figure(figsize=(12,5))

plt.plot(df['Close'], label='Historical Data')
plt.plot(
    pd.date_range(
        start=df.index[-1],
        periods=31,
        freq='D'
    )[1:],
    forecast,
    color='red',
    label='Forecast'
)

plt.legend()
plt.title("Stock Price Forecast")
plt.show()
```

Expected Deliverables

1. TSA Notebook

Include:

Dataset loading

Visualization

Moving averages

Stationarity testing

Seasonal decomposition

ARIMA forecasting

2. Forecast Plot

Final graph showing:

Historical stock prices

Future predicted prices

1. What is Stationarity?

A time series is stationary when its statistical properties (mean, variance, covariance) remain constant over time.

Examples:

White noise → stationary

Stock prices → usually non-stationary

Why important?

Most forecasting models like ARIMA assume stationary data.

2. What is Seasonal Decomposition?

Seasonal decomposition separates a time series into:

Time Series=Trend+Seasonality+Residual

Components:

Trend: Long-term movement

Seasonality: Repeating patterns

Residual: Random noise

Used to understand underlying patterns before forecasting.